Lab Reflection — AI Ethics and Sustainability

Lab: 4 **Pair:** Lena Edlind, Jakob Cederblad **Repo commit/tag:** https://github.com/leed2400/walkmates-test/commit/2943100a713fdf37e8ac90659ed4bbeda87f2020

1. What we did

Our goal was to discuss AI models such as GitHub Copilot, ChatGPT, Claude Code, and Codex, and how they can support software development. We explored their potential benefits, harms, and long-term consequences across the five dimensions of sustainability:

- Social
- Individual
- Technical
- Economic
- Environmental

We then connected software quality to broader social impacts and examined ethical questions around AI-assisted testing. To do this, we applied the concepts to the Sustainability Awareness Framework (SuSAF), identifying two chains of effects in which short-term benefits could lead to longer-term risks.

Finally, we also discussed the ethics of AI-augmented testing in greater depth, focusing on a specific trade-off and defining our stance on it.

The discussion and resulting map were documented in an analysis document.

2. What we found

We found that ethics and AI-augmented testing is a complex topic. Some aspects were difficult to place within the SuSAF framework because they affected multiple dimensions. In these cases, we decided to place each aspect in the dimension we considered to be the most affected, even though it could also influence other dimensions.

One example is the automation of repetitive tasks using AI. This can be considered both an individual and an economic aspect, as AI can support developers in their work while potentially reducing the costs for companies hiring developers. However, although this may provide short-term benefits, it could also introduce longer-term risks, such as increased dependence on AI tools or changes in the skills developers need.

This discussion emphasized the importance of considering not only the immediate benefits of AI-augmented testing but also its potential consequences for individuals, companies and society over time. It also showed that sustainability dimensions are connected, making it difficult to assess the impact of AI from a single perspective.

3. AI use (be honest — it doesn't lower your grade)

We used ChatGPT to help us organize and structure our notes for the SuSAF board. We had written our initial thoughts as continuous text without clear formatting, so we fed this raw text into the AI and asked it to format it into clearly defined "Fact" and "Effect" statements. This made it much easier to transfer our ideas onto the post-it notes.

4. Judgment

Because this assignment relies heavily on interpretation, our results might differ from other groups. If we did the assignment again, our conclusions might even shift, as we kept discovering new angles during our discussions. The placement and evaluation of different aspects within the SuSAF framework depended on our own reasoning and perspectives.

During the discussion, it was easy to focus primarily on either the positive or negative aspects of a particular finding. However, our discussions encouraged us to consider both sides and reflect on the potential trade-offs. This emphasized the importance of critically examining our own assumptions and recognising that the impact of AI-augmented testing can be interpreted differently depending on the perspectives considered.

5. What we'd test next

We would explore more chains of effects resulting from AI usage in greater depth, as well as examine the ethical considerations involved in deciding when AI should and should not be used. This would allow us to better understand the potential long-term consequences of AI-augmented testing and evaluate the trade-offs between its benefits and risks.